Example for the above procedure: Experiment: Connecting React → Express → MongoDB (Running Example)

1. Aim: To develop and run a simple MERN stack application that allows users to add names and display them using React (Frontend), Express (Backend), and MongoDB (Database).

Software Requirements

- Node.js
- MongoDB (Local Server)
- VS Code
- Chrome Browser

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Description: This application allows a user to

- Enter a name in a text field
- Click "Add" button
- Store the name in MongoDB database
- Display all stored names on the webpage

Procedure:

Step 1: Create Main Project Folder

Open command prompt and execute:

```
mkdir mern-stack
cd mern-stack
```

Step 2: Backend Setup (Express + MongoDB)

2.1 Create Server Folder

```
mkdir server
cd server
npm init -y
```

2.2 Install Required Packages

```
npm install express mongoose cors
```

2.3 Create server.js File

Create a file named **server.js** and add the following code:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());

// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/college-db")
 .then(() => console.log("MongoDB Connected"))
 .catch(err => console.log(err));

// Schema
const UserSchema = new mongoose.Schema({
 name: String
});

const User = mongoose.model("User", UserSchema);

// GET Route
app.get("/users", async (req, res) => {
 const users = await User.find();
 res.json(users);
```


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```
});

// POST Route
app.post("/users", async (req, res) => {
  const newUser = new User({ name: req.body.name });
  await newUser.save();
  res.json(newUser);
});

// Start Server
app.listen(5000, () => {
  console.log("Server running on port 5000");
});
```

2.4 Run Backend

```
node server.js
```

Output:

```
MongoDB Connected
Server running on port 5000
```

Step 3: Frontend Setup (React)

Go back to main folder:

```
cd ..
npx create-react-app client
cd client
npm install axios
```

Step 4: Connect React to Backend

Open **client/src/App.js** and replace code with:

```
import { useState, useEffect } from "react";
import axios from "axios";

function App() {
  const [name, setName] = useState("");
  const [users, setUsers] = useState([]);

  const fetchUsers = async () => {
    const res = await axios.get("http://localhost:5000/users");
    setUsers(res.data);
  };

  const addUser = async () => {
    await axios.post("http://localhost:5000/users", { name });
    setName("");
    fetchUsers();
  };

  useEffect(() => {
```


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```
  fetchUsers();
}, []);

return (
  <div style={{ padding: 20 }}>
    <h2>Simple MERN Application</h2>

    <input
      value={name}
      onChange={(e) => setName(e.target.value)}
      placeholder="Enter name"
    />

    <button onClick={addUser}>Add</button>

    <ul>
      {users.map((user) => (
        <li key={user._id}>{user.name}</li>
      ))}
    </ul>
  </div>
);
}

export default App;
```

Step 5: Run React Application

npm start

React runs on:

http://localhost:3000

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